

Headline statements, part III: the constant-Gaussian fluctuation model

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Abstract

Paper-facing wrappers for units 161 and 163 (`Laplace/Grammar/HeadlineGaussian.lean`): `headline_gaussian_dichotomy` states that for $X \sim N(0, v)$ the expected leading moment $\mathbb{E}_+[J_p(X)]$ is finite iff $\beta v < 2$, equals $J_p(0)(1 - \beta v/2)^{-p/2}$ below the threshold and $+\infty$ above it; `headline_gaussian_jensen_gap` states the strict inequality $\mathbb{E}_+[J_p(X)] > J_p(0)$ for $v > 0$. The module docstring fixes the scope (a Gaussian constant fluctuation model of the leading coefficient's positive factor; nothing about the empirical process or the fixed- n chart integral). Zero sorry/axiom.

1 Paper-facing meaning

`rem:pop_vs_emp` (ii) asserts $\mathbb{E}_{\mathcal{D}_n}[C(\xi_n)] \neq C(0)$ because the coefficients depend nonlinearly on the fluctuation. The wrappers record the two precise consequences for the simplest model: a Jensen gap and a temperature threshold $\beta v = 2$ beyond which the expected leading coefficient is infinite. The mirror `grammar_lean.tex` already carries a sentence to this effect in `rem:pop_vs_emp`; a dot is not placed since the remark is prose (bounded-statement convention).

2 Lean notes

Wrappers only; compiled on the first attempt. Separate file since the Gaussian units import `HeadlinePosterior.lean`.